

Birla Institute of Technology & Science, Pilani, Rajasthan
First Semester 2025-2026
Tutorial-1

Course: ECE/EEE F311 Communication Systems
Instructor-in-Charge: Prof. Syed Mohammad Zafaruddin

Date: 06-08-2025

Objectives

- Understand the concept of block-diagram approach, transmission bandwidth, and Low pass filtering to input signals.

Notations

$$\text{sinc}(x) = \frac{\sin(x)}{x}.$$

Tutorial Problems

1. A signal $m(t) = 8000\text{sinc}(8000\pi t)$ is passed through a channel with impulse response (i) $h(t) = 9000\text{sinc}(9000\pi t)$ (ii) $h(t) = 4000\text{sinc}(4000\pi t)$. Discuss qualitatively about the output signal for both the cases.
2. The human voice signal typically contains significant frequency components up to approximately 3.3 kHz. To model this behavior mathematically, consider a signal $m(t)$ composed of $N = 30$ harmonically related sinusoidal components with a fundamental frequency of $f_0 = 110$ Hz. The signal is defined as

$$m(t) = \sum_{k=1}^{30} \frac{1}{k} \cos(2\pi k f_0 t).$$

- (a) Compute the average power of the signal $m(t)$.
- (b) Determine a finite-duration speech segment by applying a rectangular window of width $t_0 = 10$ ms to produce the truncated signal

$$m_w(t) = m(t)w(t), \quad \text{where} \quad w(t) = \begin{cases} 1, & 0 \leq t \leq t_0, \\ 0, & \text{otherwise.} \end{cases}$$

Thus, determine the effective bandwidth of the windowed signal $m_w(t)$ by estimating the main lobe width bandwidth of the rectangular window.

- (c) A channel $h(t) = 8000\text{sinc}(8000\pi t)$ is available for transmission. Draw the transmitter block diagram.

3. In digital communication systems, information is transmitted as a stream of binary digits such as 01100011. . . . For physical transmission over a channel, each bit—'1' or '0'—must be represented by a corresponding waveform. Let bit '1' be represented by the signal $m_1(t)$ and bit '0' by $m_0(t)$. Model the digital transmission waveform by appropriately mapping the bit stream into a continuous-time signal. Assume $m_1(t)$ as a rectangular pulse of width T_1 with amplitude $+A$ and $x_0(t)$ as rectangular pulse of width T_0 with amplitude $-A$. Compute the average energy per bit E_b . Repeat the same analysis for the case where $m_1(t) = A\text{sinc}(\pi t/T_1)$ and $m_0(t) = -A\text{sinc}(\pi t/T_0)$.
4. You are transmitting a voice signal $m(t)$ to a friend in Delhi using a landline telephone system. The two telephone systems are interconnected via a twisted pair cable. Assume that the overall communication system can be modeled as a linear time-invariant (LTI) system, where the transmitter has an impulse response denoted by $g_1(t)^{\text{tx}}$, the transmission channel by $h(t)$, and the receiver by $g_2(t)^{\text{rx}}$. Mathematically express the received signal $\hat{m}(t)$ as a convolution of these components. Design an appropriate receiver filter to optimally recover the original message signal $m(t)$, and discuss the practical challenges that may arise during signal detection.

Solution

1. Using the Fourier transform pair

$$2B \operatorname{sinc}(2\pi Bt) \longleftrightarrow \Pi\left(\frac{f}{2B}\right),$$

the signal $m(t) = 8000 \operatorname{sinc}(8000\pi t)$ has a bandwidth of 4000 Hz.

- **(i):** The channel has a bandwidth of 4500 Hz, the signal's entire frequency content is preserved. The signal passes through the channel without any frequency distortion.
- **(ii):** The channel bandwidth is limited to 2000 Hz, the higher frequency components of the signal (above 2000 Hz) are attenuated. This truncation of the spectrum causes distortion in the received signal due to the loss of high-frequency content.

2. (a) To compute the average power of a continuous-time signal, we use the formula

$$P_{\text{avg}} = \frac{1}{T_0} \int_0^{T_0} x^2(t) dt.$$

For a sinusoidal signal of the form $A \cos(2\pi f_c t)$, the average power is given by $A^2/2$. In the case of the signal

$$m(t) = \sum_{k=1}^{30} \frac{1}{k} \cos(2\pi k f_0 t),$$

each harmonic component has an amplitude $1/k$, and thus contributes an average power of $1/(2k^2)$. The total average power is the sum of the powers of all harmonics:

$$P_{\text{avg}} = \sum_{k=1}^{30} \frac{1}{2k^2} = \frac{1}{2} \sum_{k=1}^{30} \frac{1}{k^2}.$$

- (b) The FT pair of the symmetric rectangular pulse:

$$\Pi\left(\frac{t}{T}\right) \leftrightarrow T \operatorname{sinc}(\pi f T)$$

The rectangular window function:

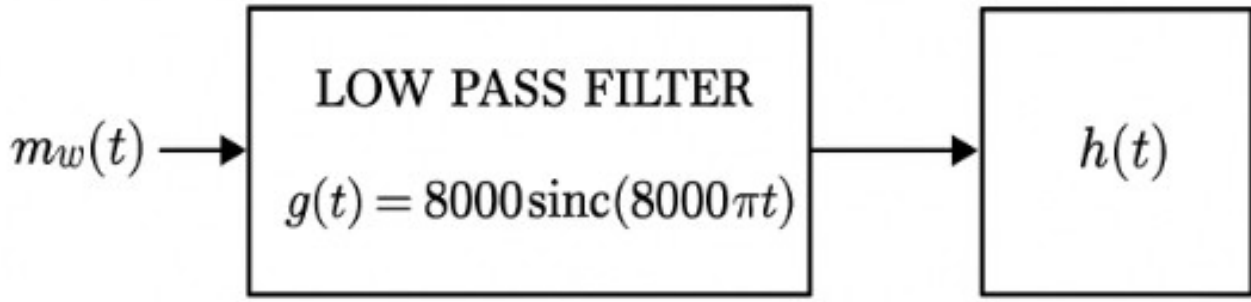
$$w(t) = \Pi\left(\frac{t - \frac{t_0}{2}}{t_0}\right)$$

FT:

$$\mathcal{F}\left\{\Pi\left(\frac{t - \frac{t_0}{2}}{t_0}\right)\right\} = t_0 \cdot \operatorname{sinc}(\pi f t_0) \cdot e^{-j\pi f t_0}$$

Thus the bandwidth of the main lobe of $W(f) = 1/t_0 = 100\text{Hz}$. The highest frequency content of $m(t)$: 3300 Hz. Multiplication in time-domain leads to convolution in the frequency domain. Thus, the BW of $m_w(t)$ is $3300 + 100 = 3400$ Hz.

- (c) The message signal $m_w(t)$, which has infinite bandwidth, is passed through a low-pass filter $g(t) = 8000 \operatorname{sinc}(8000\pi t)$ in order to limit its bandwidth and make it compatible with the channel $h(t)$.



3. Denote T_b as the bit period.

$$x(t) = \sum_{k=0}^{\infty} m_{b_k}(t - kT_b), \quad \text{where} \quad m_{b_k}(t) = \begin{cases} m_1(t), & b_k = 1 \\ m_0(t), & b_k = 0 \end{cases}$$

$$E_1 = \int_{-\infty}^{\infty} |m_1(t)|^2 dt = \int_0^{T_1} A^2 dt = A^2 T_1$$

$$E_0 = \int_{-\infty}^{\infty} |m_0(t)|^2 dt = \int_0^{T_0} A^2 dt = A^2 T_0$$

Assuming equal probability of bits '0' and '1', the average energy per bit is:

$$E_b = \frac{1}{2}E_1 + \frac{1}{2}E_0 = \frac{1}{2}A^2 T_1 + \frac{1}{2}A^2 T_0 = \frac{A^2}{2}(T_1 + T_0)$$

If $T_1 = T_0$, then $E_b = A^2$.

For the sinc function (It can be left for take home if time is limited):

$$m(t) = A \cdot \text{sinc}\left(\frac{\pi t}{T_1}\right)$$

$$M(f) = AT_1 \cdot \text{rect}(fT_1)$$

$$E_1 = \int_{-\infty}^{\infty} |M(f)|^2 df = A^2 T_1^2 \int_{-1/(2T_1)}^{1/(2T_1)} 1 df = A^2 T_1$$

Similarly

$$E_0 = \int_{-\infty}^{\infty} |M(f)|^2 df = A^2 T_0^2 \int_{-1/(2T_0)}^{1/(2T_0)} 1 df = A^2 T_0$$

Assuming equal probability of bits '0' and '1', the average energy per bit is:

$$E_b = \frac{1}{2}E_1 + \frac{1}{2}E_0 = \frac{1}{2}A^2 T_1 + \frac{1}{2}A^2 T_0 = \frac{A^2}{2}(T_1 + T_0)$$

If $T_1 = T_0$, then $E_b = A^2$.

4. Assume the voice signal $m(t)$ passes through a linear time-invariant (LTI) system comprising a transmitter $g_1(t)^{\text{tx}}$, a channel $h(t)$, and a receiver filter $g_2(t)^{\text{rx}}$. The received signal is given by the convolution:

$$\hat{m}(t) = m(t) * g_1(t)^{\text{tx}} * h(t) * g_2(t)^{\text{rx}} = m(t) * g_{\text{eq}}(t),$$

where $g_{\text{eq}}(t) = g_1(t)^{\text{tx}} * h(t) * g_2(t)^{\text{rx}}$ is the overall system impulse response.

Taking the Fourier Transform of all components, we get:

$$\hat{M}(f) = M(f) \cdot G_1(f)^{\text{tx}} \cdot H(f) \cdot G_2(f)^{\text{rx}}.$$

To recover the original message signal $m(t)$, we require $\hat{M}(f) = M(f)$, which implies:

$$G_1(f)^{\text{tx}} \cdot H(f) \cdot G_2(f)^{\text{rx}} = 1.$$

Thus, the ideal receiver filter should be designed as:

$$G_2(f)^{\text{rx}} = \frac{1}{G_1(f)^{\text{tx}} \cdot H(f)}.$$

In practice, several challenges arise:

- The channel response $H(f)$ and transmit filter $G_1(f)^{\text{tx}}$ are unknown.
- If $|H(f)|$ is small at some frequencies, inversion will be a problem.

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First Semester 2025-2026
Tutorial-2

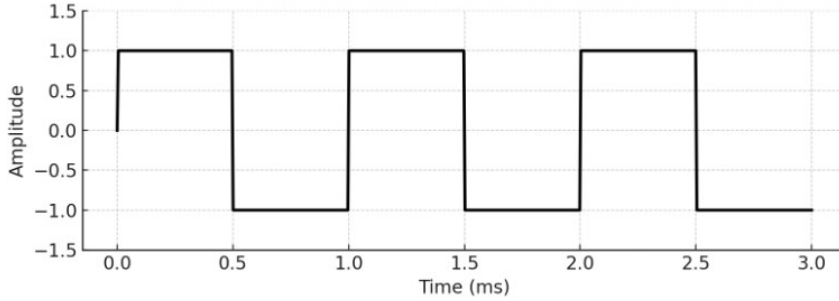
Course: ECE/EEE F311 Communication Systems
Instructor-in-Charge: Prof. Syed Mohammad Zafaruddin

Date: 13-08-2025

Objectives

- Fourier Series/Transform Applications, Distortion-less transmission, Auto-Correlation function.

Tutorial Problems



1. You are given a periodic square wave with period 1ms, as shown in figure. You also have signal processing blocks having built-in sinc function of any bandwidth. Draw a transmitter block to realize

$$x(t) = a \left[\sin(2\pi f_1 t) \right] \left[\sin(2\pi f_1 t) + \sin(2\pi f_2 t) \right] \left[\sin(2\pi f_1 t) + \sin(2\pi f_2 t) + \sin(2\pi f_3 t) \right]$$

where a is any constant with $f_1 = 1\text{KHz}$, $f_2 = 3\text{KHz}$, and $f_3 = 5\text{KHz}$. Also find the minimum channel bandwidth required for transmission.

2. *In communication systems, a channel is considered distortionless if it changes only the amplitude and timing of a signal without altering its shape. In other words, the output should be an exact replica of the input, except for a possible scaling in magnitude and a shift in time. This requires the channel to have constant gain and a phase response that varies linearly with frequency over the signal's bandwidth. Any variation in gain with frequency causes amplitude distortion, while any departure from a linear phase relationship leads to phase distortion.*

A signal $m(t)$ is transmitted through a channel of bandwidth B whose magnitude frequency response is given by

$$|H(f)| = \left[1 + k \cos(2\pi fT) \right]$$

with an additional nonlinear phase component described by

$$\theta(f) = -2\pi f t_d - k \sin(2\pi f T).$$

Derive the time-domain representation of the output signal $y(t)$ and examine the type and extent of distortion present when the parameter k is much smaller than unity ($k \ll 1$).

3. A signal

$$m(t) = 2 \cos\left(16\pi t - \frac{\pi}{4}\right) + \cos\left(40\pi t + \frac{\pi}{2}\right)$$

is transmitted through a distortionless channel. The received signal is observed to be

$$x(t) = 2 \cos\left(16\pi t + \frac{3\pi}{4}\right) + \cos(40\pi t + \phi).$$

Determine the minimum channel delay t_d and the phase ϕ of the second component.

4. Find the auto-correlation function $R_x(\tau)$ and energy/power spectral density of the following signals

- $x(t) = e^{-at}u(t)$
- $x(t) = u(t)$

Q4: if time permits.

Solution

1. The **Fourier series** for the 50% duty-cycle square wave (no derivation shown) is

$$m(t) = \frac{4A}{\pi} \left[\sin(2\pi(1 \text{ kHz})t) + \frac{1}{3} \sin(2\pi(3 \text{ kHz})t) + \frac{1}{5} \sin(2\pi(5 \text{ kHz})t) + \dots \right].$$

To construct $x(t)$:

- (a) Pass $m(t)$ through a transmitter filter $2000 \text{ sinc}(2000\pi t)$ to get the 1 kHz component, yielding $x_1(t)$.
- (b) Pass $m(t)$ through a transmitter filter $6000 \text{ sinc}(6000\pi t)$ to the signal upto 3 kHz component, yielding $x_2(t)$.
- (c) Pass $m(t)$ through a transmitter filter $10000 \text{ sinc}(10000\pi t)$ to get the signal upto 5 kHz component, yielding $x_3(t)$.

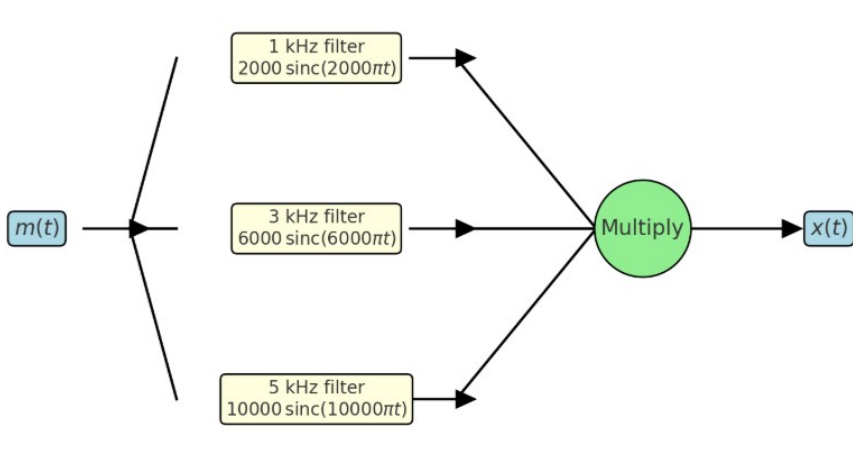
Operate all three branches in parallel, then **multiply** the outputs to obtain:

$$x(t) = x_1(t) \cdot x_2(t) \cdot x_3(t).$$

Since the multiplication produces sum-frequency terms, the **highest frequency** is:

$$f_{\max} = 1 + 3 + 5 = 9 \text{ kHz}.$$

Thus, the **minimum channel bandwidth** required is **9 kHz**.



- 2.

$$H(f) = [1 + k \cos(2\pi fT)] e^{-j2\pi f t_d} e^{-jk \sin(2\pi fT)},$$

$$\cos(2\pi fT) = \frac{1}{2} (e^{j2\pi fT} + e^{-j2\pi fT}), \quad e^{-jk \sin(2\pi fT)} = \sum_{n=-\infty}^{\infty} J_n(k) e^{-j2\pi f n T}.$$

where J_n is the Bessel function.

$$H(f) = e^{-j2\pi f t_d} \left(\sum_{n=-\infty}^{\infty} J_n(k) e^{-j2\pi f n T} \right) \left(1 + \frac{k}{2} e^{j2\pi fT} + \frac{k}{2} e^{-j2\pi fT} \right).$$

$$H(f) = e^{-j2\pi f t_d} \left[\underbrace{\sum_{n=-\infty}^{\infty} J_n(k) e^{-j2\pi f n T}}_{(A)} + \frac{k}{2} e^{j2\pi f T} \underbrace{\sum_{n=-\infty}^{\infty} J_n(k) e^{-j2\pi f n T}}_{(B)} + \frac{k}{2} e^{-j2\pi f T} \underbrace{\sum_{n=-\infty}^{\infty} J_n(k) e^{-j2\pi f n T}}_{(C)} \right]. \quad (1)$$

(A) No shift: Let $m = n$

$$\sum_n J_n(k) e^{-j2\pi f n T} = \sum_m J_m(k) e^{-j2\pi f m T}.$$

(B) $+T$ factor:

$$e^{j2\pi f T} \sum_n J_n(k) e^{-j2\pi f n T} = \sum_n J_n(k) e^{-j2\pi f (n-1) T}, \quad (2)$$

$$m = n - 1 \quad \Rightarrow \quad n = m + 1, \quad (3)$$

$$= \sum_m J_{m+1}(k) e^{-j2\pi f m T}. \quad (4)$$

(C) $-T$ factor:

$$e^{-j2\pi f T} \sum_n J_n(k) e^{-j2\pi f n T} = \sum_n J_n(k) e^{-j2\pi f (n+1) T}, \quad (5)$$

$$m = n + 1 \quad \Rightarrow \quad n = m - 1, \quad (6)$$

$$= \sum_m J_{m-1}(k) e^{-j2\pi f m T}. \quad (7)$$

Combine all terms:

$$H(f) = \sum_{m=-\infty}^{\infty} \left[J_m(k) + \frac{k}{2} (J_{m+1}(k) + J_{m-1}(k)) \right] e^{-j2\pi f (t_d + m T)}. \quad (8)$$

Using $e^{-j2\pi f a} \longleftrightarrow \delta(t - a)$,

$$h(t) = \sum_{m=-\infty}^{\infty} \left[J_m(k) + \frac{k}{2} (J_{m+1}(k) + J_{m-1}(k)) \right] \delta(t - t_d - m T)$$

Thus,

$$y(t) = m(t) * h(t) = \sum_{m=-\infty}^{\infty} \left[J_m(k) + \frac{k}{2} (J_{m+1}(k) + J_{m-1}(k)) \right] m(t - t_d - m T).$$

For $k \ll 1$, use the small-argument Bessel approximations:

$$J_0(k) \approx 1, \quad J_{\pm 1}(k) \approx \pm \frac{k}{2}, \quad J_{|m| \geq 2}(k) \approx 0.$$

$$m = 0: \quad J_0 + \frac{k}{2} (J_1 + J_{-1}) \approx 1 + \frac{k}{2} \left(\frac{k}{2} - \frac{k}{2} \right) = 1,$$

$$m = 1: \quad J_1 + \frac{k}{2} (J_2 + J_0) \approx \frac{k}{2} + \frac{k}{2} (0 + 1) = k,$$

$$m = -1: \quad J_{-1} + \frac{k}{2} (J_0 + J_{-2}) \approx -\frac{k}{2} + \frac{k}{2} (1 + 0) = 0.$$

$$y(t) \approx m(t - t_d) + k m(t - t_d - T).$$

For small k the channel introduces a post-echo of strength k at delay T .

3. Input signals:

Start time of first sinusoidal $16\pi t - \pi/4 = 0$ gives $t = 1/64$. Start time of the second sinusoidal $-1/80$.

Output signals:

Start time of first sinusoidal $16\pi t + 3\pi/4 = 2\pi$ gives $t = 5/64$. Note that using 0 instead 2π will give t negative.

Delay introduced by the channel $t_d = 5/64 - 1/64 = 1/16$.

The delay of the second component should be the same:

$$40\pi \left(-\frac{1}{80} + \frac{1}{16} \right) + \phi = 0$$

gives $\phi = -2\pi$.

4. • $x(t) = e^{-\alpha t}u(t)$.

$$X(f) = \frac{1}{\alpha + j2\pi f}$$

Energy spectral density

$$\mathcal{G}_X(f) = |X(f)|^2 = \frac{1}{\alpha^2 + 4\pi^2 f^2}.$$

Thus

$$R_X(\tau) = \mathcal{F}^{-1}[\mathcal{G}_X(f)] = \frac{1}{2\alpha} e^{-\alpha|\tau|}.$$

•

$$\begin{aligned} R_X(\tau) &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} u(t) u(t - \tau) dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{\tau}^{\frac{T}{2}} dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \left(\frac{T}{2} - \tau \right) = \frac{1}{2} \end{aligned}$$

Thus,

$$S_X(f) = \mathcal{F}[R_X(\tau)] = \frac{1}{2} \delta(f).$$

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Tutorial-3

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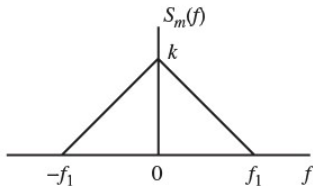
Date: 20-08-2025

Objectives

- Spectral Density, Hilbert filter, Analytical Signal.

Tutorial Problems

1. A signal $m(t)$ with power spectral density $S_m(f)$ as given in the figure is passed through a transmitter block $2f_1 \text{sinc}(2f_1\pi t)$ and then a Hilbert filter with an impulse response $\frac{1}{\pi t}$. Find the power at the output of the transmitter block and comment on the loss of the power by the transmitter block.



2. Determine the analytical signal (or complex envelope) of the signal $x(t) = \cos 2\pi f_1 t + \cos 2\pi f_2 t$, where f_1 and f_2 are positive.

Solution

1. PSD from the figure can be represented as

$$S_m(f) = \begin{cases} k \left(1 - \frac{|f|}{f_1} \right), & |f| \leq f_1, \\ 0, & \text{otherwise.} \end{cases}$$

Input power $P_m = \int_{-\infty}^{\infty} S_m(f) df = 2k \int_0^{f_1} \left(1 - \frac{f}{f_1} \right) df = 2k \left(f_1 - \frac{f_1}{2} \right) = kf_1$.

Alternatively, P_m : Area of PSD = $2 \frac{1}{2} k f_1 = kf_1$.

$$S_y(f) = |H_{LP}(f)|^2 |H_H(f)|^2 S_m(f)$$

where $|H_{LP}(f)|$ and $|H_H(f)|$ are frequency response of LPF and Hilbert filter. Using the property of Hilbert filter, $|H_H(f)|^2 = 1$ and $|H_{LP}(f)| = \text{rect}(\frac{f}{2f_1})$. Thus,

$$S_y(f) = S_m(f)$$

The output power is

$$P_y = P_m = kf_1$$

Comment: There is no loss of power.

2. The analytic signal or complex envelope of the signal is

$$x_a(t) = x(t) + jx_h(t)$$

$$x(t) = \cos(2\pi f_1 t) + \cos(2\pi f_2 t), \quad f_1, f_2 > 0,$$

$$x_h(t) = \cos(2\pi f_1 t - \pi/2) + \cos(2\pi f_2 t - \pi/2)$$

$$x_h(t) = \sin(2\pi f_1 t) + \sin(2\pi f_2 t)$$

$$x_a(t) = x(t) + jx_h(t) = \cos(2\pi f_1 t) + \cos(2\pi f_2 t) + j \sin(2\pi f_1 t) + j \sin(2\pi f_2 t) = e^{j2\pi f_1 t} + e^{j2\pi f_2 t}$$

Fourier transform:

$$X(f) = \frac{1}{2} [\delta(f - f_1) + \delta(f + f_1) + \delta(f - f_2) + \delta(f + f_2)].$$

$$X_a(f) = \delta(f - f_1) + \delta(f - f_2), \quad X_a(f) = 0 \text{ for } f < 0.$$

Main take way: Real-valued signals contain both negative and positive spectra but analytical signal has either positive/negative spectra.

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Tutorial-4

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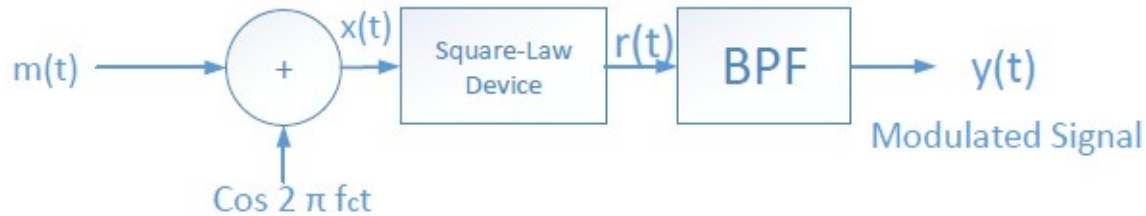
Date: 03-09-2025

Objectives

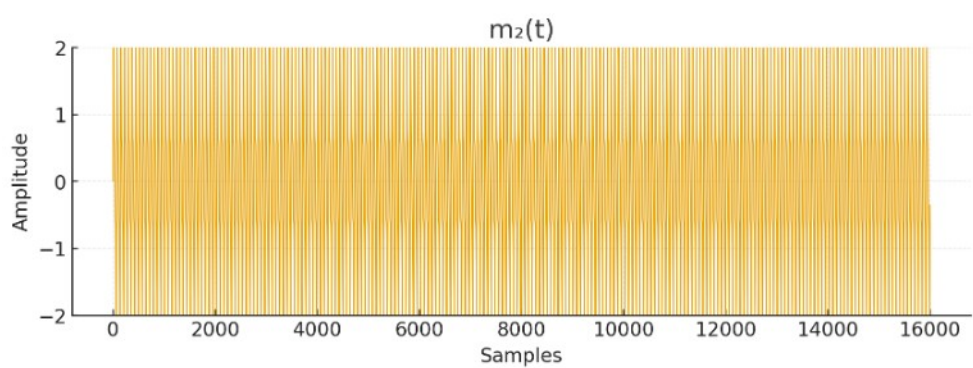
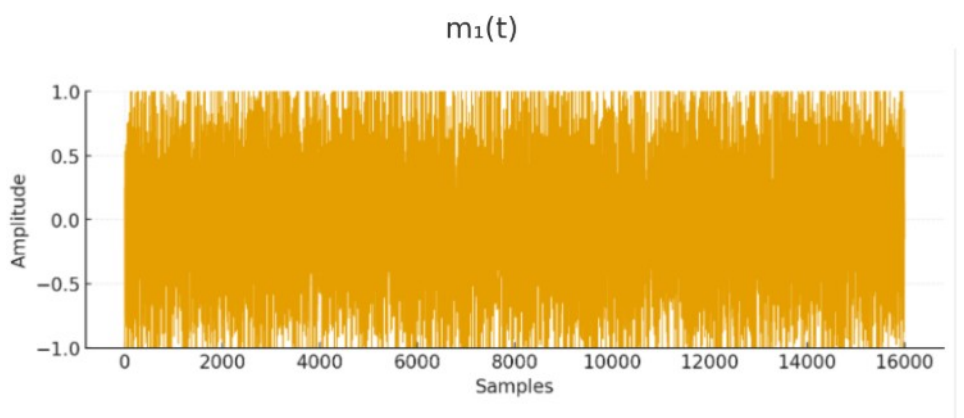
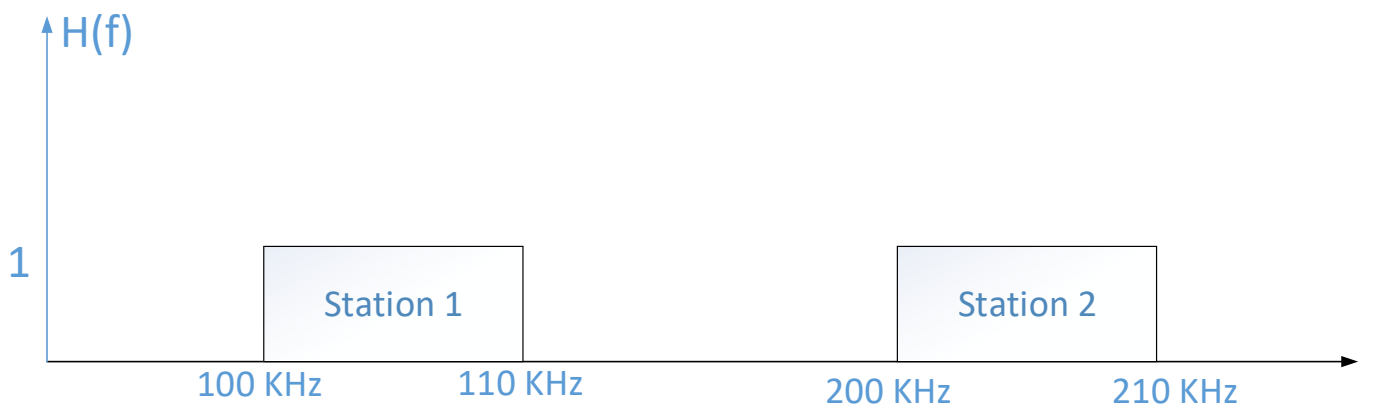
- Modulation, DSB-SC, AM

Tutorial Problems

1. Find the power efficiency for an AM signal $x(t) = \cos 220\pi t + 4 \cos 200\pi t + \cos 180\pi t$.
2. An analog passband transmission is shown in the figure below. The bandpass filter has a bandwidth $2B$ centered at f_c . The message signal $m(t)$ is a real-valued signal with amplitude varying between $-A_m$ to A_m , and has an average power P_m . The bandwidth of $m(t)$ is B . Assume $f_c \gg B$. The square-law device obeys $r(t) = 2x(t) + x^2(t)$.



- (a) Find the average transmitted power of the modulated signal $y(t)$ in terms of P_m .
 - (b) Find the modulation index of the modulated signal $y(t)$.
 - (c) Draw a system block diagram for coherent demodulation.
 - (d) What is range value of A_m that will ensure non-coherent demodulation?
3. There are two radio stations Station 1 and Station 2 transmitting audio messages $m_1(t)$ and $m_2(t)$, respectively. Available channel bandwidth and messages are shown in figures. Design both the transmitter and receiver using Amplitude Modulation (AM) with envelope detection, assuming a modulation index of 1.



Solution

1. Using

$$\cos(220\pi t) + \cos(180\pi t) = 2 \cos(200\pi t) \cos(20\pi t).$$

$$x(t) = 4 \cos(200\pi t) + 2 \cos(200\pi t) \cos(20\pi t).$$

$$x(t) = [4 + 2 \cos(20\pi t)] \cos(200\pi t).$$

Standard AM form,

$$x(t) = [A + m(t)] \cos(2\pi f_c t),$$

$$m(t) = 2 \cos(20\pi t), \quad c(t) = 4 \cos(200\pi t).$$

Carrier power $16/2 = 8$; message power: $P_m = 4/2 = 2$; Sideband signal $m(t) \cos(200\pi t)$; Sideband power $P_m/2 = 1$; Power efficiency $\eta = P_{SB}/P_{total} = 1/(1 + 8) = 1/9 = 11.1\%$

2. Signal $x(t) = m(t) + \cos 2\pi f_c t$
 $r(t) = 2x(t) + x^2(t)$

$$r(t) = 2m(t) + 2 \cos(2\pi f_c t) + m^2(t) + 2m(t) \cos(2\pi f_c t) + \cos^2(2\pi f_c t).$$

Use the trigonometric identity

$$\cos^2(2\pi f_c t) = \frac{1}{2}(1 + \cos(4\pi f_c t)),$$

$$r(t) = 2m(t) + 2 \cos(2\pi f_c t) + m^2(t) + 2m(t) \cos(2\pi f_c t) + \frac{1}{2} + \frac{1}{2} \cos(4\pi f_c t).$$

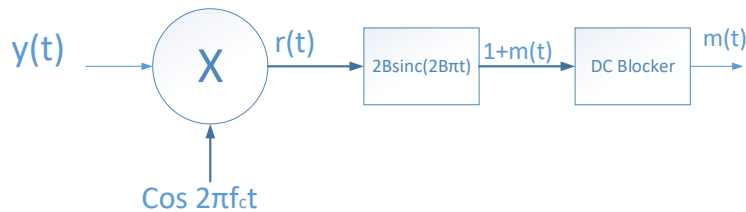
BPF at f_c will pass the signals: $2 \cos(2\pi f_c t)$ and $2m(t) \cos(2\pi f_c t)$ and suppress other terms.
 $y(t) = 2 \cos(2\pi f_c t) + 2m(t) \cos(2\pi f_c t).$

(a) $P_y = \frac{4}{2} + \frac{4P_m}{2} = 2 + 2P_m$

(b) $y(t) = [2 + 2m(t)] \cos(2\pi f_c t)$; Modulation index $\mu = \frac{m_p}{A} = \frac{2A_m}{2} = A_m$

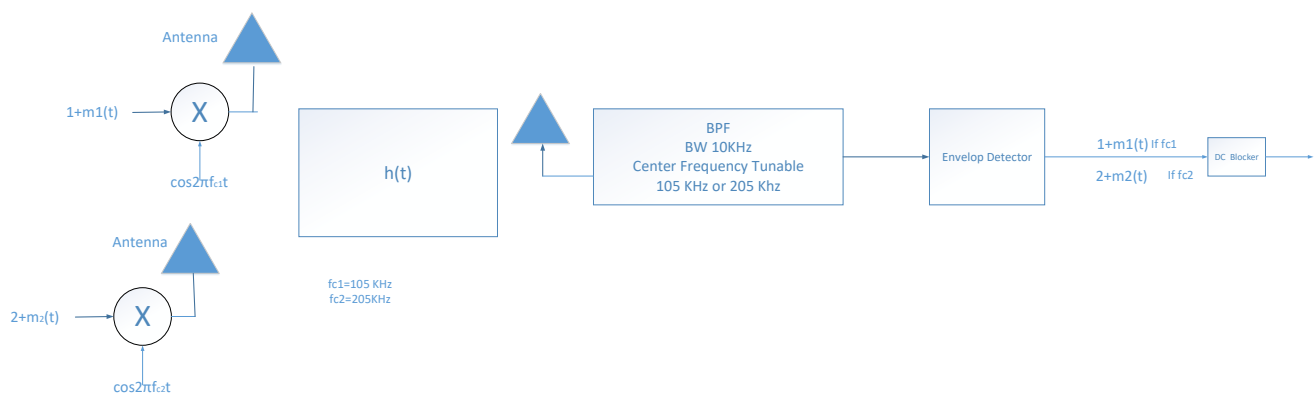
(c) Output signal:

$$r(t) = 1 + m(t) + (1 + m(t)) \cos(4\pi f_c t)$$



(d) Modulation index $\mu \leq 1$ ensure non-coherent detection. $\frac{m_p}{A} \leq 1$; $\frac{2A_m}{2} \leq 1$ gives $A_m \leq 1$.

3. Solution



Birla Institute of Technology & Science, Pilani, Rajasthan
First Semester 2025-2026
Tutorial-4

Course: ECE/EEE F311 Communication Systems
Instructor-in-Charge: Prof. Syed Mohammad Zafaruddin

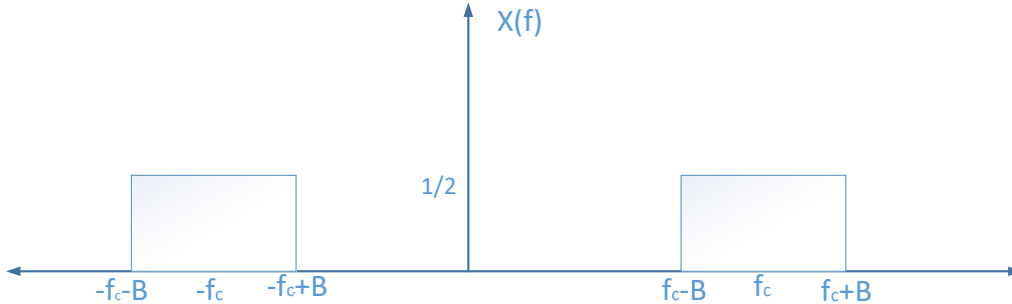
Date: 10-09-2025

Objectives

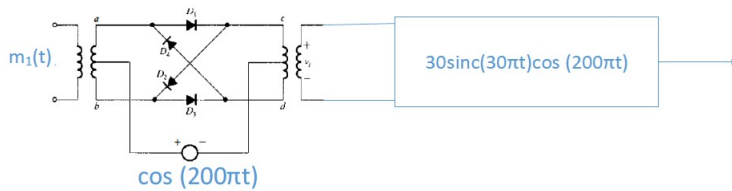
- SSB Modulation, AM Generation Circuits.

Tutorial Problems

1. A DSB-SC modulated signal at the carrier frequency f_c in the frequency domain is shown in figure below. Find the USSB in the time domain.



2. A message signal $m_1(t)$ of bandwidth 10 Hz amplitude modulates a carrier signal using the balanced ring modulator, as shown in the figure. Find the signal at the output of the bandpass filter.



Solution

1. From the DSB-SC spectrum, $m(t) = 2B \text{sinc}(2B\pi t)$
 $m(t) = \frac{\sin(2B\pi t)}{\pi t} = 2B \frac{\sin(B\pi t) \cos(B\pi t)}{B\pi t} = 2B \text{sinc}(B\pi t) \cos(B\pi t)$.
 Let $a(t)$ be *lowpass* and let $b(t)$ be *bandpass*

Then

$$H\{a(t)b(t)\} = a(t)H\{b(t)\}.$$

$$m_h(t) = \mathcal{H}\{2B \text{sinc}(B\pi t) \cos(B\pi t)\} = 2B \text{sinc}(\pi B t) \sin(\pi B t) = \frac{2 \sin^2(B\pi t)}{\pi t} = \frac{1 - \cos(2\pi B t)}{\pi t}$$

$$x_{\text{USB}}(t) = m(t) \cos(2\pi f_c t) - \hat{m}(t) \sin(2\pi f_c t) = \frac{1}{\pi t} \left[\sin(2\pi B t) \cos(2\pi f_c t) - (1 - \cos(2\pi B t)) \sin(2\pi f_c t) \right].$$

Using trigonometric identity $\sin(A + B)$ in the above

$$x_{\text{USB}}(t) = \frac{1}{\pi t} \left[\sin(2\pi(f_c + B)t) - \sin(2\pi f_c t) \right].$$

Home Work (Not to be solved): Another approach using Direct IFT:

$$S_{\text{USB}}(f) = \begin{cases} \frac{1}{2}, & f \in [f_c, f_c + B], \\ \frac{1}{2}, & f \in [-(f_c + B), -f_c], \\ 0, & \text{otherwise,} \end{cases}$$

$$\begin{aligned} s_{\text{USB}}(t) &= \int_{-\infty}^{\infty} S_{\text{USB}}(f) e^{j2\pi f t} df \\ &= 0.5 \int_{f_c}^{f_c+B} e^{j2\pi f t} df + 0.5 \int_{-(f_c+B)}^{-f_c} e^{j2\pi f t} df \\ &= 0.5 \left[\frac{e^{j2\pi(f_c+B)t} - e^{j2\pi f_c t}}{j2\pi t} \right] + 0.5 \left[\frac{e^{-j2\pi f_c t} - e^{-j2\pi(f_c+B)t}}{j2\pi t} \right] \\ &= \frac{0.5}{j2\pi t} \left(e^{j2\pi(f_c+B)t} - e^{j2\pi f_c t} + e^{-j2\pi f_c t} - e^{-j2\pi(f_c+B)t} \right) \\ &= \frac{0.5}{\pi t} \left[\sin(2\pi(f_c+B)t) - \sin(2\pi f_c t) \right]. \end{aligned}$$

$$s_{\text{USB}}(t) = \frac{1}{2\pi t} \left[\sin(2\pi(f_c + B)t) - \sin(2\pi f_c t) \right]$$

There will be a normalization factor 1/2 comparing the two approaches.

2. The ring modulator creates a switching affect by multiplying the input signal $m_1(t)$ with a square pulse train:

$$w_0(t) = \frac{4}{\pi} \left(\cos(2\pi f_c t) - \frac{1}{3} \cos(2\pi \cdot 3f_c t) + \frac{1}{5} \cos(2\pi \cdot 5f_c t) - \dots \right)$$

where $f_c = 100$ Hz. Thus, the input signal to BPF:

$$x_1(t) = w_0(t)m(t) = \frac{4}{\pi} \left(\cos(2\pi f_c t)m(t) - \frac{1}{3} \cos(2\pi \cdot 3f_c t)m(t) + \frac{1}{5} \cos(2\pi \cdot 5f_c t)m(t) - \dots \right)$$

Thus the output at the BPF is:

$$x(t) = \frac{2}{\pi} \cos(2\pi f_c t)m(t)$$

Note down the factor $1/2$.

Another switching modulator, known as the **ring modulator**, is shown in Fig. 4.6a. During the positive half-cycles of the carrier, diodes D_1 and D_3 conduct, and D_2 and D_4 are open. Hence, terminal a is connected to c , and terminal b is connected to d . During the negative half-cycles of the carrier, diodes D_1 and D_3 are open, and D_2 and D_4 are conducting, thus connecting terminal a to d and terminal b to c . Hence, the output is proportional to $m(t)$ during the positive half-cycle and to $-m(t)$ during the negative half-cycle. In effect, $m(t)$ is multiplied by a square pulse train $w_0(t)$, as shown in Fig. 4.6b. The Fourier series for $w_0(t)$ can be found by using the signal $w(t)$ of Eq. (4.5) to yield $w_0(t) = 2w(t) - 1$. Therefore, we can use the Fourier series of $w(t)$ [Eq. (4.5)] to determine the Fourier series of $w_0(t)$ as

$$w_0(t) = \frac{4}{\pi} \left(\cos \omega_c t - \frac{1}{3} \cos 3\omega_c t + \frac{1}{5} \cos 5\omega_c t - \dots \right) \quad (4.7a)$$

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Tutorial-6

Course: ECE/EEE F311 Communication Systems
Instructor-in-Charge: Prof. Syed Mohammad Zafaruddin

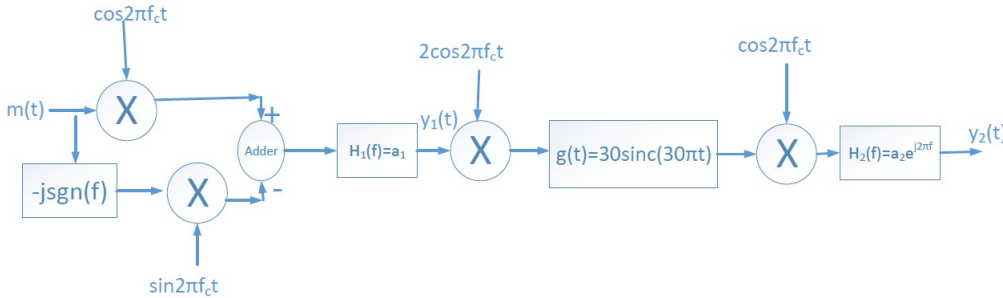
Date: 17-09-2025

Objectives

- SSB, FM

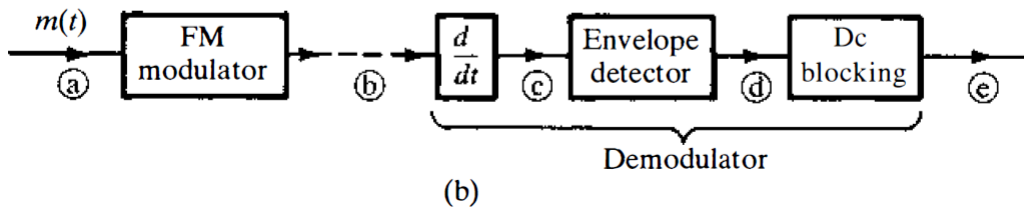
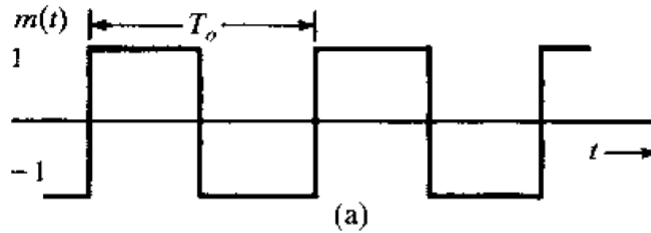
Tutorial Problems

1. A message signal $m(t)$ of bandwidth 10 Hz uses first SSB modulation over a channel of gain a_1 , then DSB-SC for a channel of gain a_2 . Find the output $y_2(t)$ in terms of the input signal $m(t)$.



2. A message signal $m(t) = 2B \text{sinc}(2B\pi t)$ frequency modulates a carrier signal $A \cos(2\pi f_c t)$ with frequency sensitivity factor k_f Hz/Volt.
 - (a) Find the integral equation of the FM signal.
 - (b) Find the BW of FM signal using Carson's rule.
3. A message signal $m(t)$ of bandwidth B phase modulates a carrier $c(t) = A \cos(2\pi f_c t)$ with a phase deviation constant k_p rad/volt. The PM modulated signal $x(t)$ is passed through a bandpass filter having a center frequency f_c Hz and bandwidth $4B$. Find the time-domain signal $y(t)$ at the output of the bandpass filter.
4. TB 5.4.2

- 5.4-2** A periodic square wave $m(t)$ (Fig. P5.4-2a) frequency-modulates a carrier of frequency $f_c = 10$ kHz with $\Delta f = 1$ kHz. The carrier amplitude is A . The resulting FM signal is demodulated, as shown in Fig. P5.4-2b by the method discussed in Sec. 5.4 (Fig. 5.12). Sketch the waveforms at points b , c , d , and e .



Solution

1. $y_1(t) = a_1(m(t) \cos 2\pi f_c t - m_h(t) \sin 2\pi f_c t)$
 $y_2(t) = a_1 a_2 (m(t-1) \cos 2\pi f_c (t-1))$
2. (a) $x(t) = A \cos(2\pi f_c t + 2\pi k_f \int_{-\infty}^t m(\tau) d\tau)$
 (b) $\text{BW} = 2(\Delta f + f_m) = 2(k_f m_p + B) = 2(2Bk_f + B)$
3. Solution is appended:

$$\begin{aligned}
 x(t) &= A \cos[2\pi f_c t + k_p m(t)] \\
 x(t) &= A \operatorname{Re} \left[e^{j2\pi f_c t + j k_p m(t)} \right] \\
 &= A \operatorname{Re} \left[e^{j k_p m(t)} e^{j2\pi f_c t} \right] \\
 &= A \operatorname{Re} \left[\left(1 + j k_p m(t) - \frac{k_p^2 m^2(t)}{2} - \frac{j k_p^3 m^3(t)}{6} + \dots \right) (\cos 2\pi f_c t + j \sin 2\pi f_c t) \right] \\
 x(t) &= A \cos 2\pi f_c t - A k_p m(t) \sin 2\pi f_c t \\
 &\quad - \frac{k_p^2 m^2(t)}{2} \cos 2\pi f_c t + \frac{k_p^3 m^3(t)}{6} \sin 2\pi f_c t + \dots
 \end{aligned}$$

First signal $\rightarrow$ An impulse in frequency domain
 Second signal $\rightarrow$ DSB-SC at f_c with BW $2B$
 Third signal $\rightarrow$ DSB-SC at f_c with BW $4B$
 Fourth signal $\rightarrow$ DSB-SC at f_c with BW $6B$
 and so on.

Thus $y(t) = \left[A \cos 2\pi f_c t - A k_p m(t) \sin 2\pi f_c t - \frac{k_p^2 m^2(t)}{2} \cos 2\pi f_c t + \dots \right]$

4. Solution is appended:

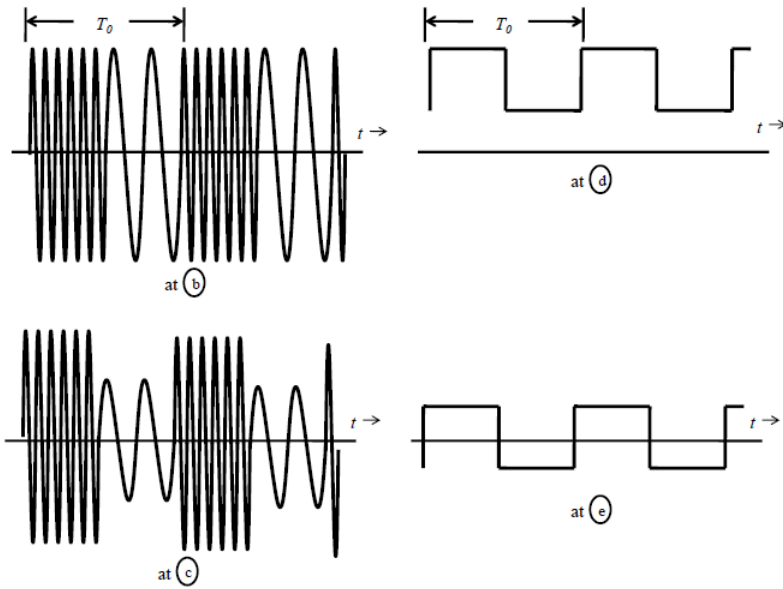
4.4-2 Given that $f_c = 10$ kHz, $\Delta f = 1$ kHz, and the message is periodic square wave of period T_0 , the resulting FM signal simply switches instantaneous frequency from 11 kHz to 9 kHz and back over one period. Thus,

$$\varphi_{\text{FM}}(t) = A \cos [20000\pi t \pm 2000\pi t]$$

over any given half period. Now, after the ideal differentiator,

$$\dot{\varphi}_{\text{FM}}(t) = -(20000\pi \pm 2,000\pi)A \sin [2,000\pi t \pm 2000\pi t]$$

Next, after the envelope detector, the output will be a periodic square wave proportional to $(20000\pi \pm 2,000\pi)A$, with a dc offset. After dc blocking, the result is a periodic square wave proportional to $m(t)$. This is illustrated in Fig. S4.4-2.



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First Semester 2025-2026
Tutorial-7

Course: ECE/EEE F311 Communication Systems
Instructor-in-Charge: Prof. Syed Mohammad Zafaruddin

Date: 25-09-2025

Objectives

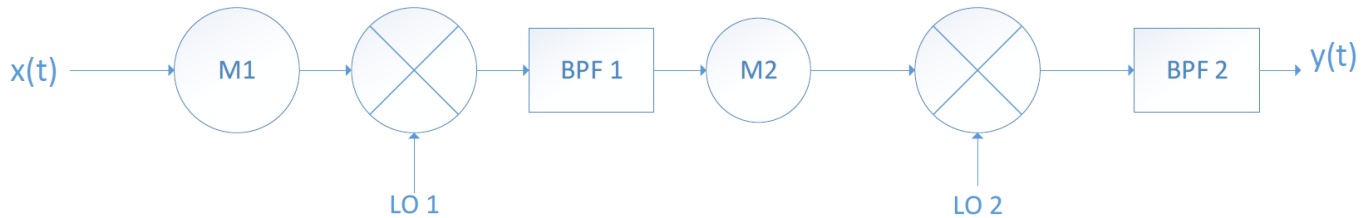
- FM Generation, Superhetrodyne Receiver

Tutorial Problems

1. An FM signal $x(t) = 20 \cos(2\pi 10^5 t + 0.3 \sin(2\pi 100 t))$ is passed through multipliers M_1 and M_2 . Local oscillators with frequencies f_{LO1} and f_{LO2} , and bandpass filters BPF 1 and BPF 2 generate another FM signal

$$y(t) = 20 \cos(2\pi 10^7 t + 15 \sin(2\pi 100 t)).$$

Find the specifications of M_1 , M_2 , f_{LO1} , f_{LO2} , and the bandwidth and carrier frequencies for BPF 1 and BPF 2. The design should satisfy the constraints: multipliers $M_1 > 1$, $M_2 > 1$, and local oscillators f_{LO1} , f_{LO2} each having a maximum frequency of 10 MHz.



2. A transmitter transmits an AM signal with a carrier frequency of 1500 kHz. When an inexpensive radio receiver (which has a poor selectivity in its RF-stage bandpass filter) is tuned to 1500 kHz, the signal is heard loud and clear. This same signal is also heard (not as well) at another dial setting. State, with reasons, at what frequency you will hear this station. The IF frequency is 455 kHz.

Solution

1. Multiple solutions depending on choice of M_1 and M_2 . We can take $M_1 = 25$ and $M_2 = 2$, which gives $\beta = 15$.

First stage:

The carrier frequency after $M_1 = 100\text{KHz} \times 25 = 2.5\text{MHz}$ and $\beta = 0.3 \times 25 = 7.5$. We take L_{O1} at 1.5MHz. Input to BPF 1: FM signal at carrier 4MHz, bandwidth of the message signal 100 Hz. Using Carson's rule $2(\beta + 1)f_m = 2(8.5) \times 100 = 1.7\text{KHz}$.

BPF 1: BW = 1.7 KHz, center frequency 4 MHz.

Second stage:

FM signal with carrier frequency (after M_2): 8 MHz with $\beta = 2 \times 7.5 = 15$ We take L_{O2} at 2 MHz to get final FM signal at 10 MHz.

BPF 2: BW = $2(15 + 1) \times 100 = 3.2$ KHz, center frequency 10 MHz.

2. Image frequency: $f_c + 2f_{IF} = 1500 + 2 \times 455 = 2410$ KHz.

Birla Institute of Technology & Science, Pilani, Rajasthan
First Semester 2025-2026
Tutorial-8

Course: ECE/EEE F311 Communication Systems
Instructor-in-Charge: Prof. Syed Mohammad Zafaruddin

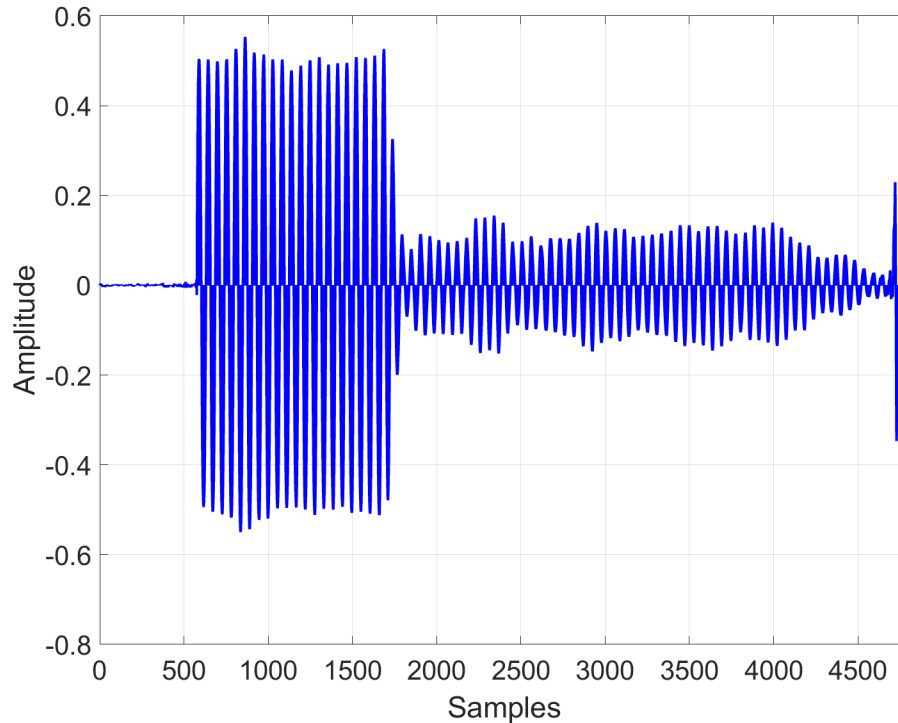
Date: 01-10-2025

Objectives

- PCM, DM, Nyquist Criteria

Tutorial Problems

1. An audio clip of duration 1 second is shown in figure. Assume average power of signal as 2 Watt and that the amplitude has signal dynamics of ± 0.6 Volt.



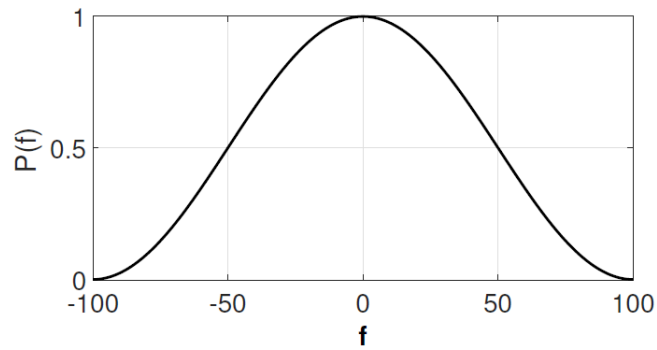
- (a) Determine the specification (i.e., number of bits/sample) of analog-to-digital converter (ADC) using PCM to achieve an SQNR of 60 dB.
- (b) Determine the minimum channel bandwidth required to transmit the audio clip with 8-ary transmission.

2. Problem from Text Book

6.7-1 In a single-integration DM system, the voice signal is sampled at a rate of 64 kHz, similar to PCM. The maximum signal amplitude is normalized as $A_{\max} = 1$.

- (a) Determine the minimum value of the step size σ to avoid slope overload.
- (b) Determine the granular noise power N_o if the voice signal bandwidth is 3.4 kHz.
- (c) Assuming that the voice signal is sinusoidal, determine S_o and the SNR.
- (d) Assuming that the voice signal amplitude is uniformly distributed in the range $(-1, 1)$, determine S_o and the SNR.
- (e) Determine the minimum transmission bandwidth.

3. Use the Nyquist frequency domain $\sum_{k=-\infty}^{\infty} P(f + kR_s) = \text{constant}$ to show that the frequency spectrum $P(f)$ of a raised cosine pulse (as shown in the figure) satisfies the Nyquist criterion for zero ISI. Find $p(t)$.



Solution

1. (a)

$$\text{SQNR}_{\text{linear}} = \frac{3L^2 P_m}{m_p^2}$$

Solving $L = 244.95$; $L = 2^N$ gives $N = 7.94$, rounded $N = 8$.

(b) From figure, assume sampling rate 4600 samples (You can take close to 4800 too as per graph) per second. $R_b = 8 \times 4600$ bps

$$R_s = \frac{8 \times 4600}{3} = 12266 \text{ symbols/sec}$$

$$B = R_s/2 = 6.133 \text{ KHz}$$

2. **Solution 4:** BP Lathi 5.7-1: Assume sinusoidal signal approximation for voice signal:

(a) $A = \frac{\Delta f_s}{2\pi f_m}$; Take $f_m = 800$ Hz. This gives $\Delta = 0.0785$

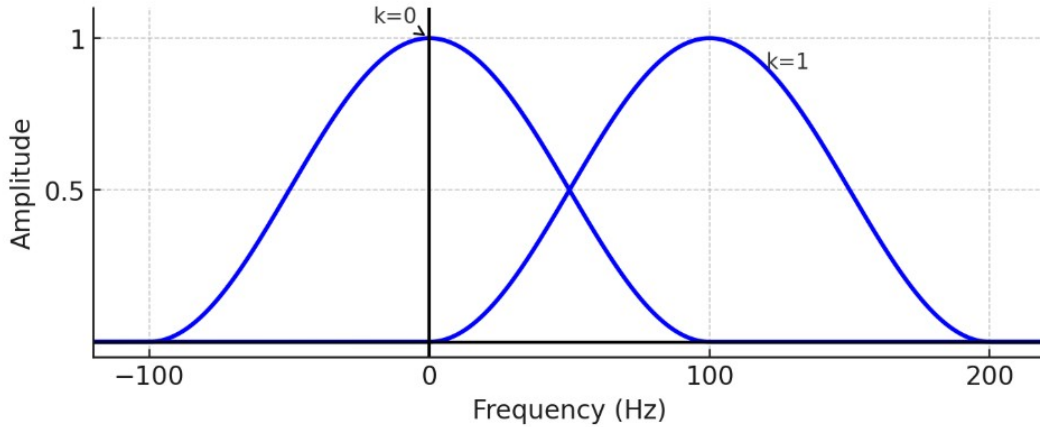
(b) $N_0 = \frac{\Delta^2 B}{3f_s} = 1.12 \times 10^{-4}$

(c) $S_0 = \frac{A^2}{2} = 0.5$; $\text{SNR} = \frac{0.5}{1.12 \times 10^{-4}} = 4.46 \times 10^3$

(d) For uniform distribution: $S_0 = \frac{m_p^2}{3}$; $\text{SNR} = \frac{0.33}{1.12 \times 10^{-4}} = 2.94 \times 10^3$

(e) BW = half of data rate = 32 KHz.

3. From the figure at $k = 0$ and $k = 1$;



$$\sum_{k=-\infty}^{\infty} P(f + kR_s) = \text{constant} \Rightarrow R_s = 100,$$

$$\text{BW} = 100 \text{ Hz},$$

$$\text{BW} = \frac{(\alpha + 1)R_s}{2} \Rightarrow \alpha = 1 \quad (\text{full raised-cosine}),$$

$$p(t) = R_s \frac{\cos(\pi R_s t)}{1 - 4R_s^2 t^2} \text{sinc}(R_s t),$$

$$p(t)|_{R_s=100} = 100 \frac{\cos(100\pi t)}{1 - 40000 t^2} \text{sinc}(100t),$$

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First Semester 2025-2026
Tutorial-8

Course: ECE/EEE F311 Communication Systems
Instructor-in-Charge: Prof. Syed Mohammad Zafaruddin

Date: 15-10-2025

Objectives

- M-ary, Random Variables

Tutorial Problems

1. Binary data is transmitted over a certain channel at a rate R_b bit/s. To reduce the transmission bandwidth, it is decided to use 8-ary PAM signaling to transmit this data.
 - (a) By what factor is the bandwidth reduced?
 - (b) By what factor is the transmitted power increased, assuming minimum separation between pulse amplitudes to be the same in both cases?
2. Solve the following:
 - (a) If the PDF of random variable X is $f_X(x) = e^{-x}$, $x > 0$, Find $P(X > 2)$.
 - (b) If the PDF of random variable X is Gaussian distributed with mean 0 and variance 1, Find $P(X > 2)$.
3. For the PDF

$$f_X(x) = \frac{1}{2\sqrt{2\pi}} e^{-x^2/32}$$

find $P(1 < x \leq 2)$

4. The noise voltage in an electronic circuit can be modeled as a Gaussian random variable $N(0, 10^{-8})$. Find the probability that the value of noise exceeds 10^{-4} .

Solution

1. Appended (BP Lathi 7.7-3):

7.7-3 (a) Each 8-ary pulse carries $\log_2 8 = 3$ bits of information. Hence, the bandwidth is reduced by a factor of 3.

(b) The amplitudes of the 8 pulses used in this 8-ary scheme are $\pm A/2$, $\pm 3A/2$, $\pm 5A/2$, and $\pm 7A/2$. Consider binary case using pulses $\pm A/2$. Let the energy of each of these pulses (of amplitude $\pm A/2$) be E_b . The power of this binary case is

$$P_{\text{binary}} = E_b R_b$$

Because the pulse energy is proportional to the square of the amplitude, the energy of a pulse $\pm \frac{kA}{2}$ is $k^2 E_b$. Hence, the average energy of the 8 pulses in the 8-ary case above is

$$E_{av} = \frac{E_b \left[2(\pm 1)^2 + 2\left(\pm \frac{3}{2}\right)^2 + 2\left(\pm \frac{5}{2}\right)^2 + 2\left(\pm \frac{7}{2}\right)^2 \right]}{8} = 21E_b$$

Hence,

$$P_{8\text{-ary}} = E_{av} \times \text{pulse rate} = 21E_b \times \frac{R_b}{3} = 7E_b R_b.$$

Therefore,

$$P_{8\text{-ary}} = 7P_{\text{binary}}$$

2. (a) $P(X > 2) = \int_2^\infty e^{-x} dx = e^{-2}.$

(b) $P(X > a) = Q(x/\sigma)$ to get $P(X > 2) = Q(2)$

3. $X \sim N(0, 4)$ with $\sigma = 2$; $P(1 < X \leq 2) = \int_1^2 f_X(x) dx = \int_1^\infty f_X(x) dx - \int_2^\infty f_X(x) dx = Q(1/\sigma) - Q(2/\sigma) = Q(1/2) - Q(2/2) = Q(0.5) - Q(1)$

4. Use $P(X > a) = Q(x/\sigma)$, where $X \sim N(0, \sigma^2)$. Here $\sigma = 10^{-4}$
 $P(X > 10^{-4}) = Q\left(\frac{10^{-4}}{10^{-4}}\right) = Q(1).$

Birla Institute of Technology & Science, Pilani, Rajasthan
First Semester 2025-2026
Tutorial-10

Course: EEE F311 Communication Systems
Instructor-in-Charge: S M Zafaruddin

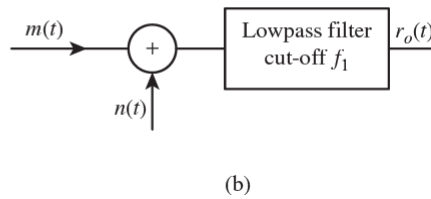
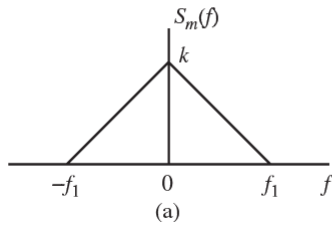
Date: 22-10-2025

1 Objectives

- Random Process

2 Tutorial Problems

1. The PSD of signal $m(t)$ is given in the figure. Find the SNR at the output if the noise PSD is $N/2$ (two-sided).



2. Find the PSD of $X(t) + Y(t)$, where $X(t)$ and $Y(t)$ are independent and $X(t) \sim U(-1, 1)$, $Y(t) \sim U(0, 1)$.
3. Q3

The random process $V(t)$ is defined as

$$V(t) = X \cos 2\pi f_c t - Y \sin 2\pi f_c t$$

where X and Y are random variables. Show that $V(t)$ is wide-sense stationary if and only if $E(X) = E(Y) = 0$, $E(X^2) = E(Y^2)$, and $E(XY) = 0$.

1. Solution is appended

(a) The lowpass filter allows the signal to pass through it but cuts off the noise spectrum beyond f_1 .

$$\text{Output signal} = r_o(t) = m_o(t) + n_o(t)$$

$$S_{mo}(f) = S_m(f)$$

and

$$S_m(f) = k \left(1 - \frac{|f|}{f_1} \right), \quad |f| \leq f_1$$

and

$$S_{no}(f) = \frac{\mathcal{N}}{2}, \quad |f| \leq f_1$$

$$\text{Therefore, output signal power} = S_o = 2k \int_0^{f_1} \left(1 - \frac{f}{f_1} \right) df = kf_1$$

The r.h.s is the area under the triangle for $S_m(f)$.

Thus, $S_o = kf_1$.

$$\text{The output noise power} = N_o = 2 \int_0^{f_1} \frac{\mathcal{N}}{2} df = \mathcal{N} f_1.$$

Hence, output SNR for case (a) is

$$(\text{SNR})_a = \frac{kf_1}{\mathcal{N}f_1} = \frac{k}{\mathcal{N}}$$

2. Solution is appended.

3. Solution

2)

$$\begin{aligned}
R_X(t + \tau, t) &= E[X(t + \tau)X(t)] = E[(X + Y)(X + Y)] \\
&= E[X^2] + E[Y^2] + E[YX] + E[XY] \\
&= E[X^2] + E[Y^2] + 2E[X]Y
\end{aligned}$$

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where the last equality follows from the independence of X and Y . But, $E[X] = 0$ since X is uniform on $[-1, 1]$ so that

$$R_X(t + \tau, t) = E[X^2] + E[Y^2] = \frac{1}{3} + \frac{1}{3} = \frac{2}{3}$$

The Fourier transform of $R_X(t + \tau, t)$ is the power spectral density of $X(t)$. Thus

$$S_X(f) = \mathcal{F}[R_X(t + \tau, t)] = \frac{2}{3}\delta(f)$$

$$\begin{aligned}
u(t) &= X \cos 2\pi ft - Y \sin 2\pi ft \\
E[u(t)] &= E(X) \cos 2\pi ft - E(Y) \sin 2\pi ft
\end{aligned}$$

and :

$$\begin{aligned}
R_{uu}(t, t + \tau) &= E\{[X \cos 2\pi ft - Y \sin 2\pi ft][X \cos 2\pi f(t + \tau) - Y \sin 2\pi f(t + \tau)]\} \\
&= E(X^2) [\cos 2\pi f(2t + \tau) + \cos 2\pi f\tau] + E(Y^2) [-\cos 2\pi f(2t + \tau) + \cos 2\pi f\tau] \\
&\quad - E(XY) \sin 2\pi f(2t + \tau)
\end{aligned}$$

For $u(t)$ to be wide-sense stationary, we must have : $E[u(t)] = \text{constant}$ and $R_{uu}(t, t + \tau) = R_{uu}(\tau)$. We note that if $E(X) = E(Y) = 0$, and $E(XY) = 0$ and $E(X^2) = E(Y^2)$, then the above requirements for WSS hold; hence these conditions are necessary. Conversely, if any of the above

Birla Institute of Technology & Science, Pilani, Rajasthan
First Semester 2025-2026
Tutorial-11

Course: EEE F311 Communication Systems
Instructor-in-Charge: S M Zafaruddin

Date: 29-10-2025

1 Objectives

- Figure of Merit, BER

2 Tutorial Problems

1. In an analog communication system, the demodulation gain is defined as the ratio of the SNR at the output of the demodulator to the SNR at the output of the noise-limiting filter at the receiver front end. Find expression for the demodulation gain of SSB.
2. Find the threshold for OOK (On-Off Keying) detection with unequal probability of transmission (p_1 for bit '1' and p_0 for bit '0') under AWGN with mean $\mu = 0$ and variance σ^2 . Take maximum signal amplitude for the pulse as A_p .

Solution

- Proven in lecture: Bandpass noise model $n(t) = n_i \cos 2\pi f_c t - n_q \sin 2\pi f_c t$
 $y(t) = Am(t) \cos 2\pi f_c t \pm Am_h(t) \sin 2\pi f_c t + n_i \cos 2\pi f_c t - n_q \sin 2\pi f_c t$
 $P_{si} = A^2 P_m$

Proven in lecture: $P_{ni} = P_{ni} = P_{nq} = P_n$

SNR at the input $\gamma_i = \frac{A^2 P_m}{P_n}$

Synchronous detector (Multiply by $\cos 2\pi f_c t$ and then LPF)

$$r(t) = \frac{A}{2}m(t) + \frac{1}{2}n_i(t)$$

$$P_{s0} = \frac{A^2}{4}P_m$$

$$P_{n0} = \frac{1}{4}P_{ni} = \frac{1}{4}P_n$$

SNR at the output $\gamma_0 = \frac{A^2 P_m}{P_n}$

Figure of Merit: $\frac{\gamma_0}{\gamma_i} = 1$

- Assume a threshold value λ .

BER for bit '1': $P_{e1} = P(A_p + n < \lambda) = Q(\frac{A_p - \lambda}{\sigma})$

BER for bit '0': $P_{e0} = P(n > \lambda) = Q(\frac{\lambda}{\sigma})$

Average BER: $P_e = p_1 P_{e1} + p_0 P_{e0}$

To get λ :

Differentiate P_e wrt to λ (Use Leibnitz rule) and to get the minima, we have

$$\frac{dP_e}{d\lambda} = p_1 \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(A_p - \lambda)^2}{2\sigma^2}} - p_0 \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(\lambda)^2}{2\sigma^2}} = 0$$

Solving above $\lambda = \frac{A_p}{2} + \frac{\sigma^2}{A_p} \log_e(\frac{p_0}{p_1})$

Course: EEE F311 Communication Systems
Instructor-in-Charge: S M Zafaruddin

Date: 12-11-2025

1 Objectives

- Matched Filter and BER

2 Tutorial Problems

1. A polar signaling system uses the following pulse in Fig. 1 for transmission with equal probability. It passes through a channel $h(t) = \delta(t)$. Take the noise PSD at the input of the matched filter as $N_0/2$. Draw the matched filter and find signal power, noise power, and BER at the output of the matched filter.

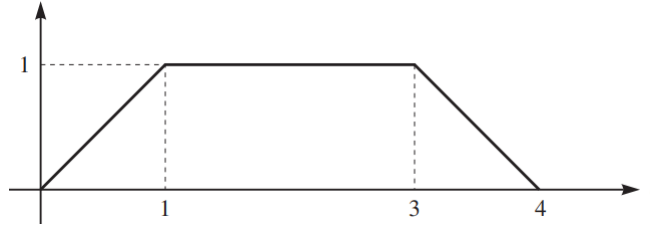


Figure 1: Problem 1

Solution: Denote the given pulse as $p(t)$.

Matched filter $f(t) = p(4 - t) = p(t)$

Amplitude of signal at the output of matched filter = Energy of $p(t)$

$$p(t) = \begin{cases} t, & 0 \leq t \leq 1 \\ 1, & 1 \leq t \leq 3 \\ -t + 4, & 3 \leq t \leq 4 \\ 0, & \text{otherwise} \end{cases}$$

$$E_p = \int_0^1 t^2 dt + \int_1^3 1^2 dt + \int_3^4 (-t + 4)^2 dt$$

$$E_p = \frac{1}{3} + 2 + \frac{1}{3} = \frac{8}{3}$$

$$\text{Amplitude} = \frac{8}{3}$$

$$\boxed{\text{Signal power} = \frac{64}{9}}$$

Noise power at the output of M.F.:

$$P_{n0} = \int_{-\infty}^{\infty} S_n(f) |F(f)|^2 df = \frac{N_0}{2} \int_{-\infty}^{\infty} |F(f)|^2 df$$

$$\Rightarrow P_{n0} = \frac{N_0}{2} \int_0^4 |f(t)|^2 dt = \frac{N_0}{2} E_p = \frac{4N_0}{3}$$

$$\boxed{P_{n0} = \frac{4N_0}{3}}$$

$$\text{BER} = Q\left(\sqrt{\frac{2E_p}{N_0}}\right) = Q\left(\sqrt{\frac{16}{3N_0}}\right)$$

2. BP Lathi Text book question 10-2-1:

10.2-1 In coherent detection of a binary PPM, a half-width pulse $p_0(t)$ of is transmitted with different delays for binary digit “0” and “1” over $0 \leq t \leq T_b$. Note that

$$p_0(t) = u(t) - u(t - T_b/2)$$

The binary PPM transmission is to simply transmit

$$\begin{aligned} & p_0(t), & \text{if “0” is sent} \\ & p_0(t - T_b/2), & \text{if “1” is sent} \end{aligned}$$

The channel noise is AWGN with spectrum level of $\mathcal{N}/2$.

- Determine the optimum receiver architecture for this binary system. Sketch the optimum receiver filter response in the time domain.
- If $P[“0”] = 0.4$ and $P[“1”] = 0.6$, find the optimum threshold and the resulting receiver bit error rate.

(a) Apply Eq. (9.22b) for $q(t) = p_0(t) = u(t) - u(t - \frac{T_b}{2})$ and $p(t) = p_0(t - \frac{T_b}{2}) = u(t - \frac{T_b}{2}) - u(t - T_b)$

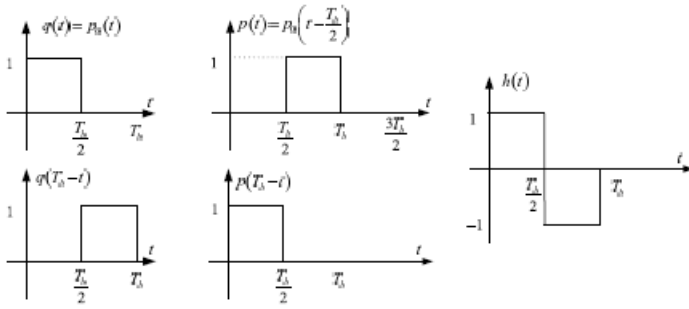


Fig. S9.2-1

$$\begin{aligned} h(t) &= \left[u(t) - u\left(t - \frac{T_b}{2}\right) \right] - \left[u\left(t - \frac{T_b}{2}\right) - u(t - T_b) \right] \\ &= u(t) - 2u\left(t - \frac{T_b}{2}\right) + u(t - T_b) \end{aligned}$$

$$\begin{aligned} E_p &= E_q = \frac{T_b}{2} \\ E_{pq} &= 0 \end{aligned}$$

Hence, the threshold is

$$a_0 = \frac{1}{2} \left(\frac{T_b}{2} - \frac{T_b}{2} \right) = 0$$

(b) The probability of error is calculated by

$$\begin{aligned} P_e &= P[0'] \cdot p_{\text{e|m}}(\epsilon|0) + P[1'] \cdot p_{\text{e|m}}(\epsilon|1) \\ &= P[0'] \cdot Q\left(\frac{a_0 - q(T_b)}{\sqrt{N/2}}\right) + P[1'] \cdot Q\left(\frac{p(T_b) - a_0}{\sqrt{N/2}}\right) \end{aligned}$$

To minimize the P_e , we have

$$\begin{aligned} \frac{\partial P_e}{\partial a_0} &= P[0'] \cdot Q'\left(\frac{a_0 - q(T_b)}{\sqrt{N/2}}\right) \sqrt{\frac{2}{N}} + P[1'] \cdot Q'\left(\frac{p(T_b) - a_0}{\sqrt{N/2}}\right) \sqrt{\frac{2}{N}} \\ &= P[0'] \cdot Q'\left(\frac{a_0 - q(T_b)}{\sqrt{N/2}}\right) \sqrt{\frac{2}{N}} + P[1'] \cdot Q'\left(\frac{p(T_b) - a_0}{\sqrt{N/2}}\right) \sqrt{\frac{2}{N}} \\ &= P[0'] \cdot \frac{1}{\sqrt{\pi N}} \exp\left(-\frac{|a_0 - q(T_b)|^2}{N}\right) \sqrt{\frac{2}{N}} - P[1'] \cdot \frac{1}{\sqrt{\pi N}} \exp\left(-\frac{|p(T_b) - a_0|^2}{N}\right) \sqrt{\frac{2}{N}} \\ &= 0 \end{aligned}$$

or

$$\begin{aligned} P[0'] \exp\left(-\frac{|a_0 - q(T_b)|^2}{N}\right) &= P[1'] \exp\left(-\frac{|p(T_b) - a_0|^2}{N}\right) \\ |a_0 - q(T_b)|^2 - |p(T_b) - a_0|^2 &= N \ln \frac{P[0']}{P[1']} \\ a_0 &= \frac{p(T_b) + q(T_b)}{2} + \frac{1}{p(T_b) - q(T_b)} \frac{N}{2} \ln \frac{P[0']}{P[1']} \\ &= \frac{1}{2} + \frac{N}{2} \ln \frac{2}{3} \end{aligned}$$

3. BP Lathi Text book question 10-6-5:

A source emits M equiprobable messages, which are assigned signals $s_1, s_2, \dots, s_M$, as shown in Fig. 2. Determine the optimum receiver and the corresponding error probability P_{eM} for an AWGN channel as a function of $E_b/\mathcal{N}$.

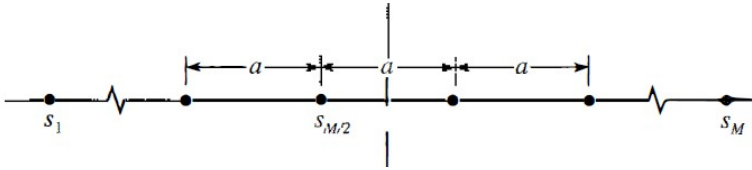


Figure 2: Problem 3

9.6-5 See Figure S9.6-5.

$$P(C|m_1) = \text{Prob}\left(n_1 < \frac{a}{2}\right) \quad \text{and} \quad P(C|m_M) = \text{Prob}\left(n_1 > \frac{-a}{2}\right)$$

$$P(C|m_2) = P(C|m_3) = \dots = P(C|m_{M-1}) = \text{Prob}\left(|n_1| < \frac{a}{2}\right)$$

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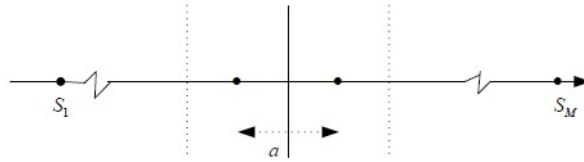


Fig. S9.6-5

Hence

$$P(C|m_1) = P(C|m_M) = \frac{1}{\sqrt{\pi\mathcal{N}}} \int_{-\infty}^{a/2} e^{-x^2/\mathcal{N}} dx = 1 - Q\left(\frac{a}{\sqrt{2\mathcal{N}}}\right)$$

$$P(C|m_2) = P(C|m_3) = \dots = P(C|m_{M-1}) = \frac{1}{\sqrt{\pi\mathcal{N}}} \int_{-a/2}^{a/2} e^{-x^2/\mathcal{N}} dx = 1 - 2Q\left(\frac{a}{\sqrt{2\mathcal{N}}}\right)$$

Hence

$$\begin{aligned} P_{eM} &= 1 - P(C) \\ &= 1 - [P(m_1)P(C|m_1) + P(m_2)P(C|m_2) + \dots + P(m_M)P(C|m_M)] \\ &= 1 - \frac{1}{M} \left[M + 2(M-1)Q\left(\frac{a}{\sqrt{2\mathcal{N}}}\right) \right] \\ &= \frac{2(M-1)}{M} Q\left(\frac{a}{\sqrt{2\mathcal{N}}}\right) \end{aligned}$$

The signal energies are

$$\left(\pm \frac{a}{2}\right)^2; \left(\pm \frac{3a}{2}\right)^2; \dots; \left(\pm \frac{M-1}{2}a\right)^2$$

Hence the average pulse energy $\overline{E}$ is

$$\begin{aligned} \overline{E} &= \frac{2}{M} \left[\frac{a^2}{4} + \frac{9a^2}{4} + \frac{25a^2}{4} + \dots + \frac{(M-1)^2 a^2}{4} \right] \\ &= \frac{a^2}{2M} \sum_{k=0}^{\frac{M-2}{2}} (2k+1)^2 = \frac{(M^2-1)}{12} a^2 \end{aligned}$$

Also

$$E_b = \frac{\overline{E}}{\log_2 M} = \frac{(M^2-1)a^2}{12 \log_2 M}$$

Hence

$$P_{eM} = \frac{2(M-1)}{M} Q\left(\sqrt{\frac{6 \log_2 M}{M^2-1} \cdot \frac{E_b}{\mathcal{N}}}\right)$$

Date: 19-11-2025

1 Objectives

- SER performance

2 Tutorial Problems

1. Compare the SER performance for 4-ary PAM and QPSK at the same E_b/N_0 at a high E_b/N_0 .
Solution: At high E_b/N_0 (write direct formula for both 4-PAM and QPSK in terms of Q-function)

$$P_e^{(4\text{-PAM})} = \frac{3}{2} Q\left(\sqrt{\frac{4}{5} \frac{E_b}{N_0}}\right)$$

$$P_e^{(\text{QPSK})} \approx 2 Q\left(\sqrt{2 \frac{E_b}{N_0}}\right)$$

Since

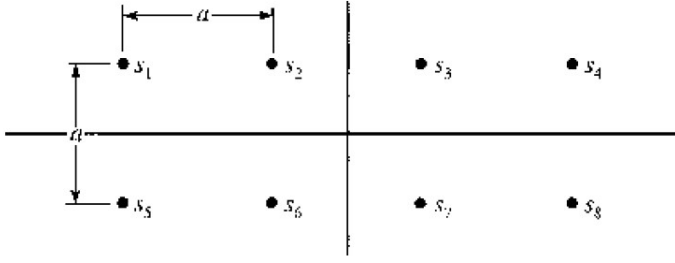
$$\sqrt{2 \frac{E_b}{N_0}} > \sqrt{\frac{4}{5} \frac{E_b}{N_0}}$$

$$\Rightarrow Q\left(\sqrt{2 \frac{E_b}{N_0}}\right) < Q\left(\sqrt{\frac{4}{5} \frac{E_b}{N_0}}\right)$$

$$P_e^{(\text{QPSK})} < P_e^{(4\text{-PAM})}$$

At high $\frac{E_b}{N_0}$, QPSK has lower P_e than 4-PAM.

2. BP Lathi Text book question 10.6-6:
Derive the SER for an 8-QAM constellation under AWGN.



$$\begin{aligned} P(C|m_1) &= P(C|m_4) = P(C|m_5) = P(C|m_8) \\ P(C|m_2) &= P(C|m_3) = P(C|m_6) = P(C|m_7) \end{aligned}$$

$$\begin{aligned} P(C|m_1) &= P\left(n_1 > \frac{a}{2}, n_2 > \frac{-a}{2}\right) \\ &= \left[1 - Q\left(\frac{a}{\sqrt{2N}}\right)\right] \cdot \left[1 - Q\left(\frac{a}{\sqrt{2N}}\right)\right] = \left[1 - Q\left(\frac{a}{\sqrt{2N}}\right)\right]^2 \\ P(C|m_2) &= P\left(|n_1| < \frac{a}{2}, n_2 > \frac{-a}{2}\right) \\ &= \left[1 - 2Q\left(\frac{a}{\sqrt{2N}}\right)\right] \cdot \left[1 - Q\left(\frac{a}{\sqrt{2N}}\right)\right] \end{aligned}$$

$$P(C) = \frac{1}{2} [P(C|m_1) + P(C|m_2)] = \frac{1}{2} \left[1 - Q\left(\frac{a}{\sqrt{2N}}\right)\right] \cdot \left[2 - 3Q\left(\frac{a}{\sqrt{2N}}\right)\right]$$

$$P_{eM} = 1 - P(C) = \frac{1}{2} Q\left(\frac{a}{\sqrt{2N}}\right) \left[5 - 3Q\left(\frac{a}{\sqrt{2N}}\right)\right]$$

In the next page, the representation of a in terms of E_b is provided. If time is short: ask the student to represent a in terms of E_b .

The average symbol power $\overline{E}$ is

$$\begin{aligned}\overline{E} &= \frac{1}{8} [4(E_{S_1} + E_{S_2})] \\ &= \frac{1}{2} \left[\frac{5a^2}{2} + \frac{a^2}{2} \right] \\ &= \frac{3a^2}{2}\end{aligned}$$

The power per bit E_b is

$$E_b = \frac{\overline{E}}{3} = \frac{a^2}{2}$$

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Therefore

$$a = \sqrt{2E_b}$$

and

$$P_{eM} = \frac{1}{2}Q\left(\sqrt{\frac{E_b}{N}}\right)\left[5 - 3Q\left(\sqrt{\frac{E_b}{N}}\right)\right]$$

Birla Institute of Technology & Science, Pilani, Rajasthan
First Semester 2025-2026
Tutorial-14

Course: EEE F311 Communication Systems
Instructor-in-Charge: S M Zafaruddin

Date: 26-11-2025

1 Objectives

- Source coding and Mutual Information

2 Tutorial Problems

1. Q1 TB 13.1-1 Appended

13.1-1 A message source generates one of four messages randomly every microsecond. The probabilities of these messages are 0.4, 0.3, 0.2, and 0.1. Each emitted message is independent of the other messages in the sequence.

- (a) What is the source entropy?
- (b) What is the rate of information generated by this source (in bits per second)?

Solution:

12.1-1 $P_1 = 0.4$, $P_2 = 0.3$, $P_3 = 0.2$, and $P_4 = 0.1$

$$H(m) = -(P_1 \log P_1 + P_2 \log P_2 + P_3 \log P_3 + P_4 \log P_4) = 1.846 \text{ bits (source entropy)}$$

There are 10^6 symbols/s. Hence, the rate of information generation is 1.846×10^6 bits/s

2. Q2 TB 13.2-1 Appended

13.2-1 A source emits seven messages with probabilities $1/2$, $1/4$, $1/8$, $1/16$, $1/32$, $1/64$, and $1/64$, respectively. Find the entropy of the source. Obtain the compact binary code and find the average length of the codeword. Determine the efficiency and the redundancy of the code.

Solution:

12.2-1

$$H(m) = \sum_{i=1}^7 P_i \log P_i = \frac{63}{32} \text{ bits}$$

$$L = \sum_i P_i L_i = \frac{1}{2}(1) + \frac{1}{4}(2) + \frac{1}{8}(3) + \frac{1}{16}(4) + \frac{1}{32}(5) + \frac{1}{64}(6) + \frac{1}{64}(6) = \frac{63}{32} \text{ binarydigits}$$

Efficiency $\eta = \frac{H(m)}{L} \times 100 = 100\%$ Redundancy $\gamma = (100 - \eta) = 0\%$. See Table S12.2-1 for the codewords.

Table S12.2-1

Message	Prob.	Code	Prob.	Code	Prob.	Code	Prob.	Code	Prob.	Code	Prob.	Code
m_1	$1/2$	0	$1/2$	0	$1/2$	0	$1/2$	0	$1/2$	0	$1/2$	0
m_2	$1/4$	10	$1/4$	10	$1/4$	10	$1/4$	10	$1/4$	10	$\rightarrow 1/2$	1
m_3	$1/8$	110	$1/8$	110	$1/8$	110	$1/8$	110	$\rightarrow 1/4$	11		
m_4	$1/16$	1110	$1/16$	1110	$1/16$	1110	$\rightarrow 1/8$	111				
m_5	$1/32$	11110	$1/32$	11110	$\rightarrow 1/16$	1111						
m_6	$1/64$	111110	$\rightarrow 1/32$	11111								
m_7	$1/64$	111111										

3. Q3 TB 13.4-1 Appended

13.4-1 A binary channel matrix is given by

$$\begin{array}{cc} & \text{Outputs} \\ & \begin{matrix} y_1 & y_2 \end{matrix} \\ \begin{matrix} \text{Inputs} \\ x_1 \\ x_2 \end{matrix} & \begin{pmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{1}{10} & \frac{9}{10} \end{pmatrix} \end{array}$$

This means $P_{y|x}(y_1|x_1) = 2/3$, $P_{y|x}(y_2|x_1) = 1/3$, etc. You are also given that $P_x(x_1) = 1/3$ and $P_x(x_2) = 2/3$. Determine $H(x)$, $H(x|y)$, $H(y)$, $H(y|x)$, and $I(x; y)$.

Solution:

12.4-1

(a)

$$P(y_1) = P(y_1|x_1)P(x_1) + P(y_1|x_2)P(x_2) = \frac{2}{3} \frac{1}{3} + \frac{1}{10} \frac{2}{3} = \frac{13}{45}$$

$$P(y_2) = 1 - P(y_1) = \frac{32}{45}$$

(b)

$$H(x) = P(x_1) \log \frac{1}{P(x_1)} + P(x_2) \log \frac{1}{P(x_2)} = \frac{1}{3} \log_2 3 + \frac{2}{3} \log_2 \frac{3}{2} = 0.918 \text{ bits}$$

To compute $H(x|y)$, we find

$$P(x_1|y_1) = \frac{P(y_1|x_1)P(x_1)}{P(y_1)} = \frac{10}{13}$$

$$P(x_1|y_2) = \frac{P(y_2|x_1)P(x_1)}{P(y_2)} = \frac{5}{32}$$

$$P(x_2|y_1) = \frac{P(y_1|x_2)P(x_2)}{P(y_1)} = \frac{3}{13}$$

$$P(x_2|y_2) = \frac{P(y_2|x_2)P(x_2)}{P(y_2)} = \frac{54}{64}$$

$$H(x|y_1) = P(x_1|y_1) \log \frac{1}{P(x_1|y_1)} + P(x_2|y_1) \log \frac{1}{P(x_2|y_1)} = \frac{10}{13} \log_2 \frac{13}{10} + \frac{3}{13} \log_2 \frac{13}{3} = 0.779$$

$$H(x|y_2) = P(x_1|y_2) \log \frac{1}{P(x_1|y_2)} + P(x_2|y_2) \log \frac{1}{P(x_2|y_2)} = \frac{5}{32} \log_2 \frac{32}{5} + \frac{54}{64} \log_2 \frac{64}{54} = 0.624$$

and

$$H(x|y) = P(y_1)H(x|y_1) + P(y_2)H(x|y_2) = \frac{13}{45}(0.779) + \frac{32}{45}(0.624) = 0.6687$$

Thus,

$$I(x|y) = H(x) - H(x|y) = 0.918 - 0.6687 = 0.2493 \text{ bits/binit}$$

$$H(y) = \sum_i P(y_i) \log \frac{1}{P(y_i)} = \frac{13}{45} \log \frac{45}{13} + \frac{32}{45} \log \frac{45}{32} = 0.8673 \text{ bits/symbol}$$

Also,

$$H(y|x) = H(y) - I(x|y) = 0.8673 - 0.2493 = 0.618 \text{ bits/symbol}$$